Lab Report: Image classification experiments

1 Introduction

In this report we compare several models on standard benchmarks. We trained each model for 20 epochs with a batch size of 64. All experiments were run on a single GPU with 16 GB of memory.

2 Results

Figure 1 shows the validation accuracy by dataset.

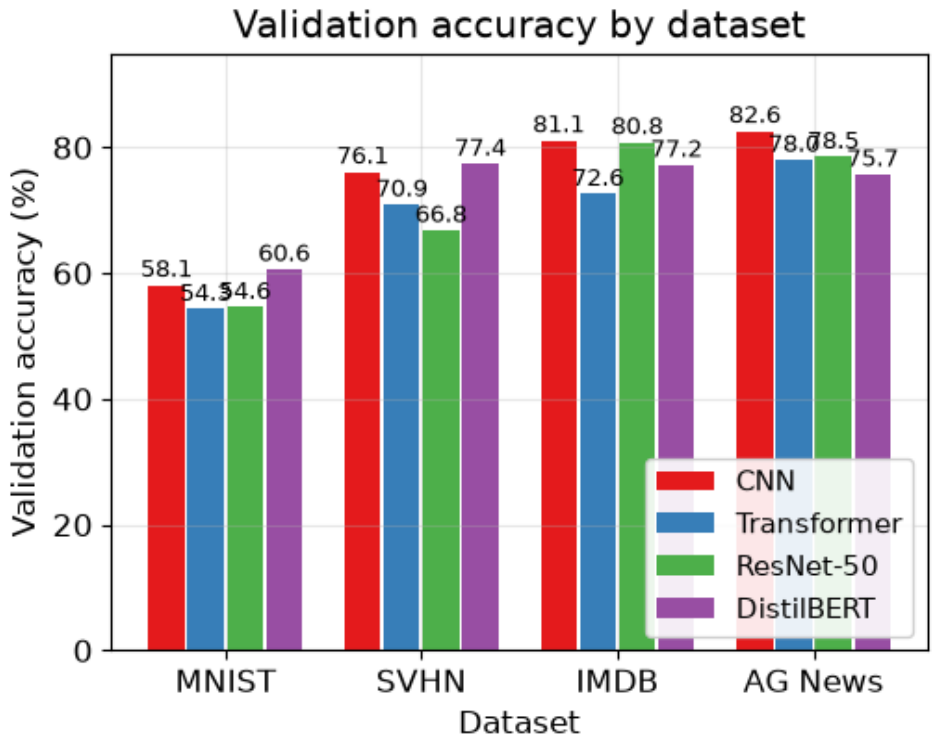


Figure 1: Validation accuracy by dataset for the models in this experiment.

Transformer achieves 76.1% validation accuracy on AG News (Figure 1). Transformer reaches 72.8% validation accuracy on SVHN. Transformer achieves over 50% validation accuracy on MNIST. On SVHN, DistilBERT is about 1 points better than CNN.

Figure 2 shows the training loss.

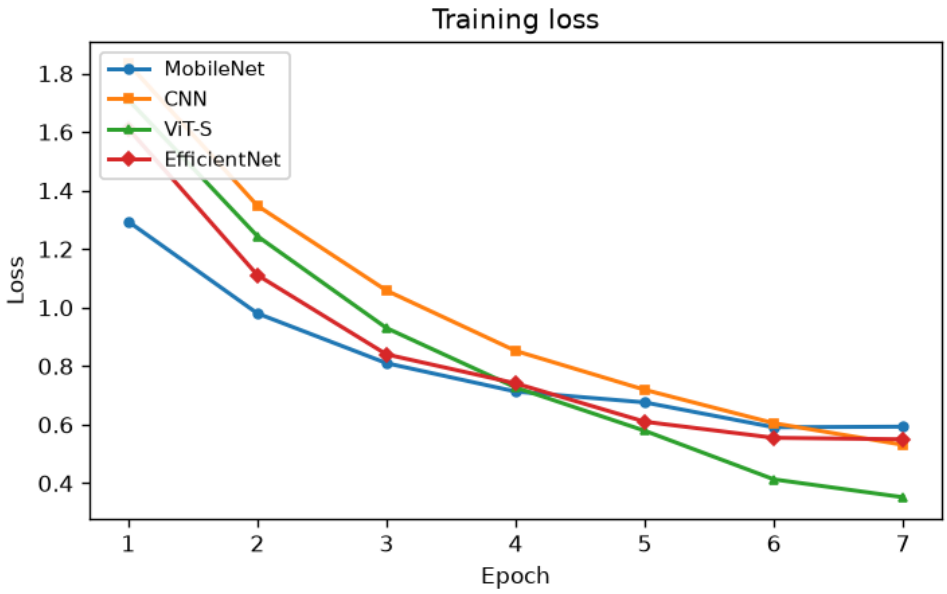


Figure 2: Training loss for the models in this experiment.

EfficientNet has a loss of 0.555 at epoch 6. At epoch 4, the loss of MobileNet is 0.852.

Figure 3 shows the training loss.

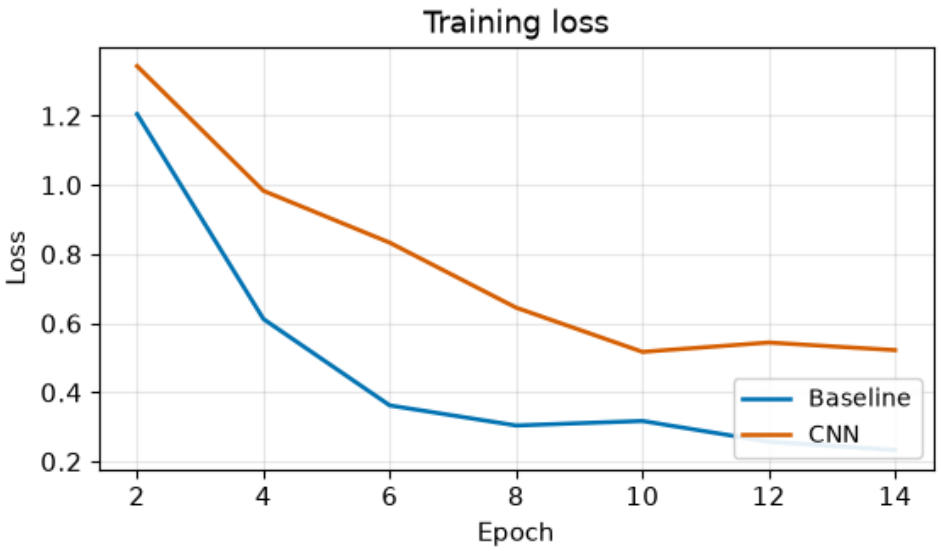


Figure 3: Training loss for the models in this experiment.

At epoch 6, the loss of Baseline is 0.36 (Figure 3). Baseline has a loss of -0.249 at epoch 6 (Figure 3). Baseline has a loss of 0.233 at epoch 14. CNN has a loss of 1.344 at epoch 2.

3 Discussion

The differences between the models are discussed in the next section.